

Lung Transplant, Pediatric

ORG: P-1300 (ISC)

[Link to Codes](#)

MCG Health

Inpatient & Surgical

Care

28th Edition

- Care Planning - Inpatient Admission and Alternatives
 - Clinical Indications for Procedure
 - Alternatives to Procedure
 - Operative Status Criteria
 - Preoperative Care Planning
- Hospitalization
 - Optimal Recovery Course
 - Goal Length of Stay - **12 days postoperative**
 - Extended Stay
 - Hospital Care Planning
- Discharge
 - Discharge Planning
 - Discharge Destination
- Evidence Summary
 - Criteria
 - Length of Stay
- References
- Footnotes
- Definitions
- Codes

Care Planning - Inpatient Admission and Alternatives

Clinical Indications for Procedure

- Procedure is indicated for **ALL** of the following(1)(2)(3)(4):[\[1\]](#)
 - ☐ Irreversible and progressive end-stage lung disease appropriate for transplant is present, as indicated by **1 or more** of the following:
 - ☐ Cystic fibrosis or non-cystic fibrosis bronchiectasis and **1 or more** of the following(1)(8)(9):
 - FEV1 less than 30% predicted
 - Rapidly declining lung function (eg, more than 30% decline over 12 months)
 - Frequent hospitalization (eg, greater than 28 days of hospitalization over the last 12 months)
 - Exacerbation requiring mechanical ventilation
 - Hypoxemia
 - Hypercapnia (PCO₂ greater than 50 mm Hg (6.7 kPa))
 - Pulmonary hypertension (eg, echocardiogram evidence of pulmonary artery systolic pressure greater than 50 mm Hg or right ventricular dysfunction)
 - Recurrent hemoptysis not controlled by embolization
 - Refractory or recurrent pneumothorax
 - Ongoing weight loss or worsening nutritional status despite aggressive nutritional supplementation
 - World Health Organization functional class IV[A]
 - ☐ Pulmonary arterial hypertension and **1 or more** of the following[B](1)(10)(11)(12):
 - High-risk pulmonary hypertension despite optimal medical therapy (eg, vasodilator, prostanoid, endothelin receptor antagonist)[C]
 - Worsening Hypoxemia
 - Progressive renal or liver dysfunction due to pulmonary hypertension and right heart failure
 - Life-threatening hemoptysis
 - Pulmonary veno-occlusive disease
 - Pulmonary capillary hemangiomatosis
 - Pulmonary vein stenosis refractory to interventional procedure
 - Congenital pulmonary disease (eg, pulmonary hypoplasia, bronchopulmonary dysplasia, alveolar capillary dysplasia, surfactant disorders, hereditary hemorrhagic telangiectasia)(1)

- ☐ Congenital heart disease and **1 or more** of the following^[D](1)(6)(13):
 - Eisenmenger physiology
 - Pulmonary hypertension
- Interstitial lung disease^[E] that is severe (eg, reduced diffusion capacity, exercise induced Hypoxemia)(14)(15)(17)
- ☐ Pulmonary Langerhans cell histiocytosis and **1 or more** of the following^[F](18):
 - Severe lung function impairment
 - Severe limitation in exercise capacity
 - Hypoxemia at rest
- ☐ Sarcoidosis with WHO functional class III^[G] or IV^[A] and **1 or more** of the following(19)(20):
 - Pulmonary hypertension unresponsive to medical treatment(21)
 - Elevated right atrial pressure greater than 15 mm Hg
 - Hypoxemia at rest
- ☐ Graft vs host disease and **1 or more** of the following(22)(23):
 - Progressive lung damage resulting in severe compromise of activities of daily living
 - Life expectancy limited by lung disease
- Other end-stage lung disease (eg, prolonged acute respiratory distress syndrome, acute fulminant lung damage including post-COVID-19 infection) not expected to recover^[H](1)(24)(25)(26)(27)
- Retransplant after failure of primary transplant (eg, chronic lung allograft dysfunction, anastomotic stenosis, bronchiolitis obliterans, restrictive allograft syndrome)(4)(28)(29)
- Alternative to lung transplant is not available or not appropriate (eg, pulmonary thromboendarterectomy for patient with chronic pulmonary thromboembolic disease).(11)(12)(30)
- Medical therapy for condition does not exist or has failed.
- No contraindications to lung transplant present

Alternatives to Procedure

- Alternatives include(1)(4)(11)(30):
 - Medical management(12)(15)
 - Atrial septostomy for patients with pulmonary arterial hypertension unresponsive to medical therapy(11)(31)
 - Left pulmonary artery to descending aorta surgical shunt ("Potts shunt") for patients with pulmonary arterial hypertension unresponsive to medical therapy(12)
 - Pulmonary thromboendarterectomy for patients with chronic pulmonary thromboembolic pulmonary hypertension(32)
 - Transcatheter embolization for arteriovenous malformations(33)
 - Lung volume reduction surgery for severe upper lobe predominant emphysema (eg, due to bronchopulmonary dysplasia)(34)
 - Palliative care(35)(36)(37)

Operative Status Criteria

Goal Length of Stay: 12 days postoperative

- Inpatient

Preoperative Care Planning

- Preoperative care planning needs may include(3)(4)(14)(38)(39)(40):
 - Preoperative evaluation, including:
 - Thorough evaluation by multidisciplinary lung transplant team for:
 - Severity of respiratory failure
 - Alternatives to lung transplant
 - Contraindications to lung transplant
 - Psychosocial evaluation including treatment adherence, mental health history, substance abuse, cognitive function, coping strategies, and social support
 - Protocol as specified by transplant center
 - Routine preoperative evaluation. See Preoperative Education, Assessment, and Planning Tool [SR](#).
 - Diagnostic test scheduling, including:
 - Right-sided heart catheterization
 - Pulmonary function testing
 - 6-minute walk test
 - Cardiopulmonary exercise test with measurement of maximum oxygen uptake (VO₂ maximum)
 - Echocardiogram(41)
 - Chest radiographic studies(42)
 - Quantitative ventilation/perfusion lung scan(43)
 - Estimation of lung allograft size
 - Cardiac MRI (eg, for connective tissue disease or sarcoidosis with suspected myocardial involvement)(39)

- Laboratory testing
- Hematologic evaluation for coagulopathy and thrombophilia (hypercoagulable state) (eg, for patient with lupus or other connective tissue disease)(14)(39)
- Nutrition evaluation(8)(44)
- Evaluation for gastroesophageal reflux and aspiration risk (eg, esophageal manometry, pH monitoring, barium swallow)(14)(39)
- Dental evaluation
- Immunology screening, including blood and tissue typing
- Detailed infectious disease screening, including(8)(45)(46):
 - Sputum cultures or bronchoscopy to rule out bacterial colonization with virulent multidrug-resistant organisms
 - Viral antibody titers, including HIV, cytomegalovirus, hepatitis B and C, varicella-zoster
 - Viral antigen testing for hepatitis B, Epstein-Barr virus
 - HIV, hepatitis B, and hepatitis C nucleic acid testing
 - Nasopharyngeal PCR for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)
 - Purified protein derivative (PPD) or interferon gamma release assay (eg, Quantiferon TB gold or T-spot.TB)
 - Screening for regional organisms as appropriate (eg, *Histoplasma* species in central United States, *Coccidioides* species in southeastern United States, *Blastomyces* species in midwestern, southern United States)(46)
- Preoperative treatment, procedures, and stabilization, including addressing all treatable coexisting conditions before transplant
- Vaccinations (eg, influenza; hepatitis; Streptococcus pneumoniae; SARS-CoV-2; measles, mumps, and rubella; Herpes varicella-zoster; Rotavirus)(46)
- Preoperative pulmonary rehabilitation
- Smoking cessation counseling
- Extracorporeal membrane oxygenation support as a bridge to transplant as indicated(1)(47)(48)(49)(50)
- Mechanical ventilation as a bridge to transplant in selected cases(49)(50)
- Preoperative discharge planning as appropriate. See Discharge Planning in this guideline.

Hospitalization

Optimal Recovery Course

Day	Level of Care	Clinical Status	Activity	Routes	Interventions	Medications
1	- OR to ICU - Social Determinants of Health Assessment - Discharge planning	Clinical Indications met^[I] - Intubated	Bed rest	- IV fluids - IV medications - NPO	- Central lines - ECG - Oximetry - Oxygen - Laboratory tests - CXR postoperatively - Chest tube	- Perioperative antibiotics - Immunosuppressive medications and corticosteroids - Possible inhaled or parenteral vasoactive medication - Multimodal analgesia, possible PCA or epidural analgesics
2	- ICU - Social Determinants of Health Assessment	Hypotension absent	Bed rest	- Possible enteral feeds^[J] - IV fluids - IV medications	- Oximetry - Central lines - Oxygen - Portable CXR - Chest tube	- Perioperative antibiotics - Immunosuppressive medications and corticosteroids - Possible inhaled or parenteral vasoactive medication - Multimodal analgesia
3	- ICU - Social Determinants of Health Assessment	No evidence of myocardial ischemia Dangerous arrhythmia absent	Bed rest	- Possible enteral feeds - IV fluids - IV medications	- Oximetry - Oxygen - Central lines - Possible chest tube	Epidural analgesics absent - Antibiotics absent - Immunosuppressive medications and corticosteroids - Possible inhaled or parenteral vasoactive

						- medication - Multimodal analgesia
4	- ICU - Social Determinants of Health Assessment	Hemodynamic stability	Bed rest	- Possible enteral feeds - IV fluids - IV medications	- Oximetry - Oxygen - Central lines - Possible chest tube	Wean inhaled or parenteral vasoactive medication PCA absent - Immunosuppressive medications and corticosteroids - Oral or IV analgesic agent
5	- ICU - Social Determinants of Health Assessment	Mechanical ventilation absent	Bed rest	- Possible enteral feeds - IV fluids - IV medications	Chest tube absent - Oxygen - Oximetry - Central lines	Inhaled or parenteral vasoactive medication absent - Immunosuppressive medications and corticosteroids - Oral or IV analgesic agent
6	- Possible intermediate care - Social Determinants of Health Assessment	Pain absent or managed	Dangle legs[K]	Begin oral diet - IV fluids - IV medications	- Oxygen - Oximetry - Possible central lines	IV analgesic agent absent - Immunosuppressive medications and corticosteroids - Oral analgesic agent as needed
7	- Intermediate care or floor - Social Determinants of Health Assessment	Pulmonary edema absent or mild	Up to chair[K]	- Advance diet - IV fluids - IV medications	Central lines absent - Oxygen - Oximetry	- Immunosuppressive medications and corticosteroids - Oral analgesic agent as needed
8	- Intermediate care or floor - Social Determinants of Health Assessment	Rejection absent or controlled Stable fluid balance	Limited ambulation[K]	Oral medications - Advance diet - IV fluids or oral hydration	- Oxygen - Oximetry	- Immunosuppressive medications and corticosteroids - Oral analgesic agent as needed
9	- Intermediate care or floor - Social Determinants of Health Assessment	SaO₂ greater than 90% on room air	Limited ambulation[K]	Oral hydration Oral medications - Advance diet	Taper oxygen - Oximetry	- Immunosuppressive medications and corticosteroids - Oral analgesic agent as needed
10	- Floor - Social Determinants of Health Assessment	Oxygenation stable with activity	Limited ambulation[K]	Oral hydration Oral medications Oral diet	Oxygen absent - Oximetry	- Immunosuppressive medications and corticosteroids - Oral analgesic agent as needed

11	- Floor - Social Determinants of Health Assessment	No evidence of postoperative or surgical site infection	Ambulatory[L]	- Oral hydration - Oral medications - Oral diet	Oximetry	- Immunosuppressive medications and corticosteroids - Oral analgesic agent as needed
12	- Floor - Social Determinants of Health Assessment	Lung function studies acceptable	Ambulatory	- Oral hydration - Oral medications - Oral diet	Oximetry	Immunosuppressive medication regimen for next level of care initiated - Oral analgesic agent as needed
13	- Social Determinants of Health Assessment - Floor to discharge - Complete discharge planning	Hemodynamic stability Rejection absent No evidence of postoperative or surgical site infection SaO₂ greater than 90% on room air and stable with activity Pain absent or managed Discharge plans and education understood	Ambulatory or acceptable for next level of care[L]	Oral hydration[M] Oral medications or regimen acceptable for next level of care Oral diet or acceptable for next level of care	Oximetry	Immunosuppressive medication regimen for next level of care established - Oral analgesic agent as needed

(3)(4)(50)(52)(53)[N](#)

Recovery Milestones are indicated in **bold**.

Goal Length of Stay: 12 days postoperative

Note: Goal Length of Stay assumes optimal recovery, decision making, and care. Patients may be discharged to a lower level of care (either later than or sooner than the goal) when it is appropriate for their clinical status and care needs.

Extended Stay

Minimal (a few hours to 1 day), Brief (1 to 3 days), Moderate (4 to 7 days), and Prolonged (more than 7 days).

- Extended stay beyond goal length of stay may be needed for(42)(50)(52)(53)(54):
 - Failure to meet discharge criteria (recovery milestones within final day in Optimal Recovery Course)
 - Expect brief stay extension.
 - Primary graft dysfunction(50)(52)(55)(56):
 - Anticipate continued mechanical ventilation.
 - Anticipate use of inhaled nitric oxide or epoprostenol for 2 to 5 days for severe hypoxemia or pulmonary hypertension.
 - Anticipate possible extracorporeal membrane oxygen support until lung function recovers.(49)
 - Expect moderate to prolonged stay extension.
 - Aspiration pneumonitis
 - Anticipate bedside swallow evaluation and possible barium swallow or functional endoscopic evaluation of swallowing (FEES).
 - Expect brief stay extension.
 - Infection(46)(57)
 - Anticipate intravenous antifungal, antiviral, or antibiotic agents.
 - Anticipate diagnostic testing including CT chest, cultures, viral and fungal serology, galactomannan and beta-D glycan levels.

- Pneumonia may require bronchoscopy for bronchoalveolar lavage, transbronchial biopsy, and protected brush cultures.
 - Expect moderate stay extension.
- Postoperative empyema(58)
 - Anticipate cultures, chest tube drainage, antibiotic treatment, and possible thoracoscopic decortication.
 - Expect moderate to prolonged stay extension.
- Acute or hyperacute rejection(59)
 - Anticipate increased immunosuppression, plasma exchange, bronchoscopy, and lung biopsy.
 - Expect moderate stay extension.
- Surgical complications(54)(58)
 - Hemorrhage, pneumothorax, hemothorax, empyema, or persistent air leaks may require prolonged chest tube drainage.
 - Hemothorax may require correction of coagulopathy, transfusion, and possible surgical exploration.
 - Chylothorax may require dietary modification, thoracic duct embolization, or reoperation for thoracic duct ligation.
 - Expect brief to moderate stay extension.
- Recurrent pleural effusion(58)
 - Recurrent effusions may require thoracentesis, catheter drainage, or thoracoscopic intervention.
 - Expect brief to moderate stay extension.
- Heart-lung transplant(6)(7)
 - Complex surgery and complications such as phrenic or vagus nerve dysfunction, gastroparesis, and chylothorax may require longer inpatient care.
 - Anticipate longer dependence on mechanical ventilation.
 - Expect brief to moderate stay extension.
- Lung-liver transplant(7)(60)
 - Complications from liver transplant (eg, hepatic artery or portal vein thrombosis, graft nonfunction, liver hematoma) may require longer inpatient care.
 - Post liver transplant graft reperfusion syndrome may cause hypotension and compromise lung function.
 - Vasopressor support and conservative fluid management required for lung allograft may compromise liver graft recovery.
 - Expect moderate stay extension.
- Bronchial anastomotic defects, necrosis, or dehiscence(54)(61)
 - Anticipate possible bronchoscopic intervention (eg, stent, cyanoacrylate glue) or surgical revision of anastomosis.
 - Surgical intervention (eg, drainage, fistula closure, reanastomosis, or flap bronchoplasty) may be required.
 - Expect brief to moderate stay extension.
- Pulmonary artery anastomotic complication (eg, thrombosis, kinking, torsion)(58)
 - Expect possible reoperation and revision of anastomosis.
 - Expect moderate to prolonged stay extension.
- Venous thromboembolism(62)
 - Anticipate CT angiogram and anticoagulation.
 - Expect brief to moderate stay extension.
- Pulmonary venous thrombosis
 - Anticipate CT or MR angiography.
 - Expect brief to moderate stay extension.
- Combined lung transplant with repair of underlying congenital heart lesion
 - Complex surgery and complications such as phrenic nerve dysfunction may require longer inpatient care.
 - Expect moderate to prolonged stay extension.
- Need for prolonged postoperative mechanical ventilation(52)
 - Cardiopulmonary bypass may be required for patient with severe pulmonary hypertension, concomitant cardiac procedure, or inability to tolerate single-lung ventilation or pulmonary artery clamping during transplant.
 - Anticipate prolonged postoperative mechanical ventilation and higher risk of graft dysfunction.
 - Expect moderate stay extension.
- Acute kidney injury(63)
 - Anticipate careful fluid management with balanced crystalloid solutions, adjustment of nephrotoxic agents, and possible renal replacement therapy.
 - Expect brief stay extension.
- Care for comorbidities
 - Patient with active comorbidities, such as chronic heart failure, liver disease, chronic renal disease, or diabetes mellitus may require continued inpatient care.
 - Expect brief to moderate stay extension.
- Postoperative atrial tachyarrhythmia(64)
 - Anticipate rate control (eg, beta-blocker) or antiarrhythmic agent.
 - Expect brief to moderate stay extension.
- High preoperative lung allocation score(65)
 - Anticipate preoperative hospital and ICU days for stabilization of organ function.
 - Anticipate higher incidence of infection, Acute kidney injury (stage 2), Acute renal failure (stage 3 acute kidney injury), and need for inpatient dialysis.

- Expect moderate to prolonged stay extension.

See Common Complications and Conditions [↗](#) ISC for further information.

Hospital Care Planning

- Hospital evaluation and care needs may include(3)(4)(52):
 - Diagnostic test scheduling and completion, including:
 - CBC, chemistries monitoring
 - Chest CT scan and other chest radiographic studies as indicated(42)(66)
 - Transesophageal echocardiogram
 - Pulmonary function testing
 - Lower extremity Duplex ultrasound for routine deep venous thrombosis screening(67)
 - Treatment and procedure scheduling and completion, including(2):
 - Steroids
 - IV antibiotics(46)
 - Prophylactic anti-infective regimen, including antifungal, anti-CMV, and anti-*Pneumocystis jirovecii* agents(46)
 - Bronchoscopy(68)
 - Early extubation
 - Thoracentesis
 - Establishment of immunosuppression protocols(69)
 - Inhaled pulmonary vasodilator (eg, nitric oxide, epoprostenol) for pulmonary hypertension, right ventricular failure, or refractory hypoxia
 - Extracorporeal membrane oxygenation[N](49)
 - Consultation, assessment, and other services scheduling and completion, including:
 - Medical and surgical specialty consultations, as indicated
 - Dietitian(51)
 - Monitoring patient's status for deterioration and comorbid conditions (see Inpatient Monitoring and Assessment Tool [↗](#) SR); key items include:
 - Hemodynamic stability measurements, which may initially include:
 - Blood pressure via arterial line
 - Pulmonary artery pressure via right-sided heart catheter
 - Central venous pressure
 - Cardiac output and cardiac index
 - Urine output
 - Monitoring for arrhythmias
 - Pleural drainage quantity and appearance and need for chest tubes
 - Pulmonary status, including presence of:
 - Dyspnea or cough
 - Hypoxemia
 - Hypercarbia
 - Signs and symptoms of respiratory failure or diaphragm dysfunction
 - Wound tenderness, swelling, or serous drainage from incision
 - Pain management
 - Renal function
 - Gastrointestinal status
 - Glycemic control (eg, due to high-dose corticosteroids)
 - Signs and symptoms of infection or acute rejection, including:
 - Low-grade fever
 - Elevated WBC count
 - Decrease in FEV₁ more than 10%
 - New infiltrates on chest x-ray
 - Pleuritic chest pain
 - New or increased sputum production
 - Purulent drainage from site of infection
 - Increasing oxygen requirements

Discharge

Discharge Planning

- Discharge planning includes[O]:
 - Assessment of needs and planning for care, including(71)(72):
 - Develop and modify treatment plan (involving multiple providers) as needed.

- Evaluate and address preadmission functioning as needed.
- Evaluate and address psychosocial status issues as indicated. See Psychosocial Assessment [SR](#) for further information.
- Evaluate and address social determinants of health (eg, housing, food). See Social Determinants of Health Screening Tool [SR](#) for further information.(70)
- Evaluate and address patient or caregiver preferences as indicated.
- Identify skilled services needed at next level of care, with specific attention to:
 - Cardiovascular status assessment(2)
 - Medication management, adherence instruction, and side effects assessment(2)(73)(74)
 - Pain management(75)
 - Respiratory status assessment(2)
 - Wound or dressing management(76)
- Early identification of anticipated discharge destination; options include(72)(77):
 - Home, considerations include:
 - Home safety assessment. See Home Safety Assessment [SR](#) for further information.
 - ☐ Patient safe to go home; examples include(78)(79)(80):
 - Medical status stable for patient's condition
 - Functional care can safely be provided with available resources.
 - Mental status stable for patient's condition
 - Medication availability confirmed and reconciliation complete
 - Patient/caregiver education completed with written discharge instructions provided
 - Community resources identified and referrals made, as needed
 - Home care arranged, if indicated
 - Necessary medical equipment delivery arranged or available in home, if indicated
 - Necessary medical supplies ordered, or patient/caregiver can obtain, if indicated
 - Access to follow-up care
 - Self-management ability if appropriate. See Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment [SR](#) for further information.
 - Caregiver need, ability, and availability
 - Post-acute skilled care or custodial care as indicated. See Discharge Planning Tool [SR](#) for further information.
- Transitions of care plan complete, including(72)(77)(81):
 - Patient and caregiver education complete. See Lung Transplant, Pediatric: Patient Education for Clinicians [SR](#) for further information.
 - See Teach Back Tool [SR](#) for further information.
 - ☐ Medication reconciliation completion includes(74)(82):
 - Compare patient's discharge list of medications (prescribed and over-the-counter) against provider's admission or transfer orders.
 - Assess each medication for correlation to disease state or medical condition.
 - Report medication discrepancies to prescribing provider, attending physician, and primary care provider, and ensure accurate medication order is identified.
 - Provide reconciled medication list to all treating providers.
 - Confirm that patient or caregiver can acquire medication.
 - Educate patient and caregiver.
 - Provide complete medication list to patient and caregiver.
 - Importance of presenting personal medication list to all providers at each care transition, including all provider appointments
 - Reason, dosage, and timing of medication (eg, use "teach-back" techniques)(83)
 - Encourage communication between patient, caregiver, and pharmacy for obtaining prescriptions, setting up home medication delivery, and reviewing for drug-drug interactions.
 - See Medication Reconciliation Tool [SR](#) for further information.
 - Plan communicated to patient, caregiver, and all members of care team, including(84):
 - Inpatient care and service providers
 - Primary care provider
 - All post-discharge care and service providers
 - Appointments planned or scheduled, which may include:
 - Primary care provider
 - Specialists for management of comorbidities as needed(85)
 - Thoracic surgeon
 - Transplant center or team(86)
 - Other
 - Outpatient testing and procedure plans made, which may include:

- Laboratory testing(85)
- Other
- Referrals made for assistance or support, which may include:
 - Community services
 - Educational program (eg, chronic condition management, self-management)(87)
 - Financial, for follow-up care, medication, and transportation
 - Self-help or support groups(88)
 - Tobacco use treatment
 - Other
- Medical equipment and supplies coordinated (ie, delivered or delivery confirmed), which may include:
 - Blood pressure cuff
 - Child safety seat(89)
 - Nutritional supplements
 - Weight scale
 - Wound care equipment and supplies(90)
 - Other

Discharge Destination

- Post-hospital levels of admission may include:
 - Home.
 - Home healthcare. See Home Care Indications for Admission Section [HC](#) in Transplant, Pediatric guideline in Home Care.

Evidence Summary

Criteria

The evidence for the clinical indications found in this guideline includes 23 published peer reviewed articles, 6 specialty society or other evidence-based guidelines, and 1 book section.

A multivariate analysis of 930 pediatric lung transplants found that adult donor lung allografts did not negatively impact survival.(5) **(EG 2)** Heart-lung transplant may be performed in patients with certain congenital heart diseases and Eisenmenger syndrome not amenable to surgical correction or medical therapy.(6)(7) **(EG 2)**

Length of Stay

Analysis of national hospital discharge data for a pediatric population shows 26% of hospitalized patients undergoing the principal procedure of lung transplant discharged in 12 days or less.(53) **(EG 3)**

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Footnotes

[A] World Health Organization functional class IV describes patients with pulmonary hypertension who have signs of right-sided heart failure, symptoms of dyspnea at rest, and more severe symptoms with activity. [A in Context Link 1, 2, 3]

[B] Patients with pulmonary hypertension may be candidates for combined heart-lung transplant. Specific criteria include World Health Organization functional class IV symptoms, a cardiac index less than 2 L/min/m², and a mean right atrial pressure above 15 mm Hg.(1) [B in Context Link 1]

[C] The International Society of Heart and Lung Transplantation (ISHLT) classifies pulmonary hypertension as "high risk" based on the European Pediatric Pulmonary Disease Network score, which includes points for right ventricular failure, symptom progression, syncope, failure to thrive, World Health Organization functional class III or IV, rising NT-proBNP level, and hemodynamic severity indicators from echocardiogram and right heart catheterization data.(1)(10) [C in Context Link 1]

[D] Heart-lung transplant may be performed for certain congenital heart diseases, or for Eisenmenger syndrome not amenable to surgical correction or medical therapy.(1)(6) [D in Context Link 1]

[E] Interstitial lung disease refers to a category of noninfectious nonmalignant diseases of the lung parenchyma characterized by inflammation, fibrosis, and destruction of the alveoli. In children, these disorders are more commonly due to inhalation injury or toxin exposure, surfactant disorders, hypersensitivity pneumonitis, cryptogenic organizing pneumonia, lung involvement from connective tissue disease and other known causes, whereas idiopathic pulmonary fibrosis is more common in adults and rarely occurs in children.(14)(15) (16) [E in Context Link 1]

[F] Langerhans cell histiocytosis is also known as histiocytosis X and eosinophilic granuloma. [F in Context Link 1]

[G] World Health Organization functional class III describes patients who have significant physical activity limitation due to pulmonary hypertension. Although asymptomatic at rest, they develop symptoms of dyspnea with less than ordinary activity. [G in Context Link 1, 2]

[H] The latest International Society of Heart and Lung Transplantation guideline has no formal recommendations regarding lung transplant criteria for patients with fulminant lung damage caused by COVID-19 infection; however, broad criteria that have been proposed by the ISHLT and applied at transplant centers include radiographic and clinical evidence of severe irreversible disease after allowing at least 28 days for possible recovery, absence of current infection (eg, 2 negative PCR tests by endotracheal aspirate or bronchoscopy 24 to 48 hours apart), single organ failure, and otherwise appropriate for transplant.(1)(24)(25) [H in Context Link 1]

[I] See Clinical Indications for Procedure in this guideline. [I in Context Link 1]

[J] Enteral feeding and oral intake may be delayed in patients with connective tissue disease who have gastrointestinal involvement.(51) [J in Context Link 1]

[K] For very young patients (eg, infants), activity goals should be modified based on age and development. [K in Context Link 1, 2, 3, 4, 5]

[L] Patient is ambulatory or near baseline activity for age and development. [L in Context Link 1, 2]

[M] Some patients may have their hydration and nutritional needs met via alternative means (eg, percutaneous endoscopic gastrostomy or jejunostomy tube), especially those with connective tissue diseases that pose a high risk of postoperative aspiration.(51) [M in Context Link 1]

[N] Indications for postoperative extracorporeal membrane oxygenation include pulmonary hypertension with right ventricular dysfunction, primary graft dysfunction, or high risk for developing primary graft dysfunction (eg, lobar lung transplant, donor size discrepancy).(49) [N in Context Link 1]

[O] Discharge instructions should be given in the patient's and caregiver's native language using trained language interpreters whenever possible.(70) [O in Context Link 1]

Definitions

Acute kidney injury (stage 2)

- Acute kidney injury (stage 2) with significant clinical findings, as indicated by **ALL** of the following(1)(2)(3):
 - Acute[A] kidney injury (stage 2), as indicated by **1 or more** of the following[B]:
 - Within past 7 days, rise in serum creatinine between 2 and 2.9 times its baseline value (increase of 3 times baseline would qualify as stage 3 acute kidney injury)
 - Decreased urine output,[C] as indicated by **ALL** of the following:
 - Assessment of urine output appropriate (eg, adequate volume status, no urinary obstruction)
 - Urine output less than 0.5 mL/kg/hr for 12 hours
 - Significant clinical findings, as indicated by **1 or more** of the following[B]:
 - Hemodynamic instability
 - Worsening clinical status despite observation care (eg, rising creatinine or urine output remains less than 0.5 mL/kg/hr)
 - Altered mental status

- Cardiac arrhythmias of immediate concern
- Severe hypertension (SBP greater than 180 mm Hg or DBP greater than 120 mm Hg in adults or greater than the 95th percentile for age, gender, and height plus 30 mm Hg in pediatric patients)(4)(5)
- Significant respiratory findings (eg, pulmonary edema) that are severe (eg, Hypoxemia) or persist despite observation care
- Clinically significant electrolyte abnormality that is severe (eg, hyperkalemia with severe ECG findings)[D] or persists despite observation care
- Clinically significant metabolic abnormality (eg, metabolic acidosis) that is severe or persists despite observation care

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Footnotes

- A. A specialty society practice guideline defines acute kidney injury as a change in renal function that is known or presumed to have occurred over the previous 7 days.(1)
- B. In determining the stage of acute kidney injury (eg, stage 2 vs stage 3), patients should be staged according to the criteria that give them the highest stage.(1)
- C. The use of urine output as an indicator of acute kidney injury is important, as change in urine output can occur before a noticeable rise in creatinine.(1)(3) It is also suggested that urine output can identify patients with acute kidney injury that may otherwise be missed.(1)(3) However, urine output as a metric is only valid under the appropriate conditions. For example, the patient must not be hypovolemic, there must not be urinary obstruction, and the collection and measurement of urinary output must be complete and accurate.(1)(3) Use of urinary output to define acute kidney injury can be inaccurate in obese patients due to the nonlinear relationship between body weight and urine output. For example, in a 100-kg (220-lb) patient, urine output of 45 mL/hour over 12 hours (adequate output) would qualify as stage 2 acute kidney injury.(1)(3)
- D. ECG findings include peaked T-waves, PR interval greater than 0.20 seconds, QRS interval greater than 0.12 seconds, and arrhythmia.(6)(7)

Acute renal failure (stage 3 acute kidney injury)

- Acute[A] kidney injury (stage 3),[B] as indicated by **1 or more** of the following[C](1)(2)(3):
 - Within past 7 days, rise in serum creatinine to 3 times its baseline value or higher
 - Within past 7 days, rise in serum creatinine to at least 1.5 times its baseline value (increase of 50%) and serum creatinine is greater than 4 mg/dL (354 micromoles/L)
 - In child younger than 18 years, estimated glomerular filtration rate less than 35 mL/min/1.73m² (0.59 mL/sec/1.73m²)
-  eGFR - Pediatric Calculator and **1 or more** of the following:
 - Within past 7 days, rise in serum creatinine to at least 1.5 times its baseline value (increase of 50%)
 - Within past 48 hours, rise in serum creatinine greater than 0.3 mg/dL (26.5 micromoles/L)
- Decreased urine output,[D] as indicated by **ALL** of the following:
 - Assessment of urine output appropriate (eg, adequate volume status, no urinary obstruction)
 - Oligoanuria, as indicated by **1 or more** of the following:
 - Urine output less than 0.3 mL/kg/hour for 24 hours
 - Anuria (urine output less than 0.1 mL/kg/hour) for 12 hours
- Initiation of renal replacement therapy (RRT) required(4)

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Footnotes

- A. A specialty society practice guideline defines acute kidney injury as a change in renal function that is known or presumed to have occurred over the previous 7 days.(1)
- B. For purposes of these criteria, the term acute renal failure can be viewed as stage 3 acute kidney injury in the KDIGO classification system.(1) Of note, in the ICD-10 diagnostic coding system (main coding scheme used in labeling diagnoses), the relevant codes covering the clinical entity of significant acute kidney injury use the term acute kidney failure (eg, code N17.9).
- C. In determining the stage of acute kidney injury (eg, stage 2 vs stage 3), patients should be staged according to the criteria that give them the highest stage.(1)
- D. The use of urine output as an indicator of acute kidney injury is important, as change in urine output can occur before a noticeable rise in creatinine.(1)(3) It is also suggested that urine output can identify patients with acute kidney injury that may otherwise be missed.(1)(3) However, urine output as a metric is only valid under the appropriate conditions. For example, the patient must not be hypovolemic, there must not be urinary obstruction, and the collection and measurement of urinary output must be complete and accurate.(1)(3) Use of urinary output to define acute kidney injury can be inaccurate in obese patients due to the nonlinear relationship between body weight and urine output. For example, in a 100-kg (220-lb) patient, urine output of 45 mL/hour over 12 hours (adequate output) would qualify as stage 2 acute kidney injury.(1)(3)

Altered mental status

- Altered mental status (ie, different from baseline), as indicated by **1 or more** of the following(1)(2)(3)(4):
 - Confusional state (eg, disorientation, difficulty following commands, deficit in attention)
 - Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment)
 - Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
 - Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
 - Coma (ie, not arousable)

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Cardiac arrhythmias of immediate concern

- Cardiac arrhythmias or findings of immediate concern, as indicated by **1 or more** of the following(1)(2)(3):
 - Heart rhythms that are inherently dangerous or unstable, as indicated by **1 or more** of the following(4)(5)(6):
 - Resuscitated ventricular fibrillation or cardiac arrest
 - Ventricular escape rhythm
 - Sustained ventricular tachycardia (30 seconds or more of ventricular rhythm at greater than 100 beats per minute)
 - Nonsustained ventricular tachycardia and **1 or more** of the following:
 - Suspected cardiac ischemia as cause or consequence of ventricular tachycardia
 - Acute myocarditis
 - Unstable cardiac conduction defects, as indicated by **1 or more** of the following(6)(7)(8):
 - Type II second-degree atrioventricular block
 - Third-degree atrioventricular block
 - New-onset left bundle branch block with suspected myocardial ischemia
 - Any heart rhythm and **1 or more** of the following(4)(5)(9)(10)(11):
 - Continuous long-term ECG monitoring needed (eg, initiation of drug requiring monitoring beyond observation care)
 - Patient has automatic implanted cardioverter-defibrillator that is repeatedly firing, malfunctioning, or in need of immediate adjustment of settings beyond scope of observation care.
 - Heart rhythms of concern due to **1 or more** of the following:

- Hypotension
- Respiratory distress
- Association with other significant symptoms (eg, syncope)(9)(10)(12)

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Dangerous arrhythmia

- Dangerous arrhythmia, as indicated by **1 or more** of the following(1)(2)(3):
 - Heart rhythms that are inherently dangerous or unstable, as indicated by **1 or more** of the following(4)(5)(6):
 - Resuscitated ventricular fibrillation or cardiac arrest
 - Ventricular escape rhythm
 - Sustained (30 seconds or more of ventricular rhythm at greater than 100 beats per minute) ventricular tachycardia
 - Nonsustained ventricular tachycardia and **1 or more** of the following:
 - Acute myocarditis
 - Myocardial ischemia
 - Unstable cardiac conduction defects, as indicated by **1 or more** of the following(6)(7)(8):
 - Type II second-degree atrioventricular block
 - Third-degree atrioventricular block
 - New-onset left bundle branch block with suspected myocardial ischemia
 - Heart rhythms of concern due to **1 or more** of the following:
 - Hypotension
 - Respiratory distress
 - Association with other significant symptoms (eg, syncope)(9)(10)(11)

References

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11. Westerman S, Manogue M, Hoskins MH. Bradycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:996-1002.

Hemodynamic instability

- Hemodynamic instability, as indicated by **1 or more** of the following(1)(2)(3)(4)(5)(6):
 - Vital sign abnormality not readily corrected by appropriate treatment, as indicated by **1 or more** of the following[A]:
 - Tachycardia that persists despite appropriate treatment (eg, volume repletion, treatment of pain, treatment of underlying cause)
 - Hypotension that persists despite appropriate treatment (eg, volume repletion, treatment of underlying cause)
 - Orthostatic hypotension that persists despite appropriate treatment (eg, volume repletion)
 - Hypotension that is severe, as indicated by **ALL** of the following:
 - Hypotension
 - Severe hypotension, as indicated by **1 or more** of the following:
 - Lactate of 2.0 mmol/L (18 mg/dL) or more secondary to hypotension (ie, hypoperfusion)[B](3)(6)
 - Metabolic acidosis (arterial or venous[C] pH less than 7.35) not otherwise explained
 - Mean arterial pressure[A][D] less than 65 mm Hg
 - IV inotropic or vasopressor medication required to maintain adequate blood pressure or perfusion

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Footnotes

- A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. There are numerous causes of an elevated lactate level. The most common are cardiogenic or hypovolemic shock, severe heart failure, severe trauma, or sepsis. However, there are also other etiologies to consider such as vigorous exercise, seizures, liver disease, or medication use (eg, metformin, beta2-agonists); therefore, interpretation of elevated lactate levels requires consideration of the clinical context (eg, well-appearing post exercise vs hypotensive). The severity and persistence of the elevation can sometimes be helpful in this differentiation. In most instances of lactic acidosis, blood pH is less than 7.35, with a serum bicarbonate level 20 mEq/L (mmol/L) or lower. However, a coexisting respiratory or metabolic alkalosis can mask these findings.(6)(7)(8)(9)(10)
- C. Analyses have concluded that venous and arterial pH measurements are highly correlated, with small differences between them that are usually not clinically significant.(11)(12)(13)(14)(15) If a mixed acid-base disorder is suspected, an arterial blood gas is indicated. (12)
- D. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as mean arterial pressure (MAP) equals 1/3 SBP + 2/3 DBP.(1)(4) A calculator is available at <https://www.mdcalc.com/mean-arterial-pressure-map>.

Hemodynamic stability

- Hemodynamic stability, as indicated by **1 or more** of the following:
 - Hemodynamic abnormalities at baseline or acceptable for next level of care
 - Patient hemodynamically stable, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
 - Tachycardia absent
 - Hypotension absent
 - No evidence of inadequate perfusion (eg, no myocardial ischemia)
 - No other hemodynamic abnormalities (eg, no Orthostatic hypotension)

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Hypotension

- Hypotension, as indicated by **ALL** of the following(1)(2)(3)(4):
 - Not patient baseline (eg, healthy adult with low SBP) or intentional therapeutic goal (eg, low SBP as treatment goal in heart failure)
 - Low blood pressure, as indicated by **1 or more** of the following:
 - SBP less than 90 mm Hg in adult or child 10 years or older[A]
 - Mean arterial pressure[A][B] less than 70 mm Hg in adult or child 10 years or older

- Decrease in baseline mean arterial pressure of 25% or more,^{[A][B]} with significant signs or symptoms due to lower blood pressure (eg, near syncope, syncope, chest pain)
- SBP less than sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age^[A]
- SBP less than 70 mm Hg in infant 1 to 11 months of age^[A]

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Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = $1/3 \text{ SBP} + 2/3 \text{ DBP}$. A calculator is available at <https://www.mdcalc.com/mean-arterial-pressure-map>.

Hypotension absent

- Hypotension absent^[A], as indicated by **1 or more** of the following⁽¹⁾⁽²⁾⁽³⁾⁽⁴⁾:
 - SBP greater than or equal to 90 mm Hg in adult or child 10 years or older
 - Mean arterial pressure^[B] greater than or equal to 70 mm Hg in adult or child 10 years or older
 - Mean arterial pressure^[B] at patient's baseline (eg, healthy adult with low SBP), at intentional therapeutic goal (eg, patient with heart failure), or acceptable for next level of care (eg, blood pressure stable and no significant signs or symptoms due to low blood pressure)
 - SBP greater than or equal to sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age
 - SBP greater than or equal to 70 mm Hg in infant 1 to 11 months of age

References

1. Jones D, Di Francesco L. Hypotension. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:657-64.
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4. Singh S, Holmes JF. Pediatric trauma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:2052-2066.e3.

Footnotes

- A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = $1/3 \text{ SBP} + 2/3 \text{ DBP}$.

Hypoxemia

- Hypoxemia, as indicated by **1 or more** of the following⁽¹⁾:
 - Patient without baseline need for supplemental oxygen with oxygen saturation (SaO₂) of less than 90% or arterial blood gas partial pressure of oxygen (PO₂) of less than 60 mm Hg (8.0 kPa) on room air^{[A][B]}

- Patient without baseline need for supplemental oxygen who now requires supplemental oxygen to keep SaO₂ greater than 89% or PO₂ greater than 59 mm Hg (7.9 kPa)[A]
- Patient with baseline need for supplemental oxygen who now requires increased supplemental oxygen to maintain oxygenation at baseline or acceptable level

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Footnotes

- A. Pulse oximetry (SpO₂) may underestimate arterial saturation (SaO₂), especially in people with dark skin pigmentation, thick skin, cold body temperature, or who are wearing colored nail polish, and thus may not accurately detect hypoxemia.(2) Where appropriate, pulse oximetry should be measured after warming fingers or removing nail polish. Pulse oximetry results should be assessed within the context of the patient's clinical presentation, and performance of an arterial blood gas considered for a more accurate assessment of oxygenation if indicated.(2) Specific target values may require adjustment when care is delivered at high altitude.(1)(3)
- B. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Multimodal analgesia

- Multimodal analgesia involves the utilization of 2 or more analgesic agents with different mechanisms of action in order to provide additive or synergistic pain control, while minimizing side effects and reliance on opioids.(1)(2)(3)

References

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No contraindications to lung transplant present

- No contraindications to lung transplant present, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
 - No malignancy that has high risk of recurrence or cancer-related mortality[A]
 - No untreatable failure or dysfunction of another major organ system unless combined organ transplants can be performed[B]
 - No stroke, acute coronary syndrome, or myocardial infarction within 30 days[C](1)
 - No incurable extrapulmonary infection
 - No active *Mycobacterium tuberculosis* infection
 - No HIV infection with detectable viral load[D](1)
 - No uncontrolled systemic (extrapulmonary) disease[E](7)
 - No bilateral diaphragmatic paralysis or severe diaphragmatic weakness(7)
 - No severe tracheobronchomalacia
 - No severe transpleural systemic to bronchial artery collaterals or aorto-pulmonary collaterals
 - BMI not over 35  BMI Calculator
 - No uncorrectable bleeding diathesis
 - No active alcohol, tobacco, or illicit substance misuse or dependence[F](1)
 - No unaddressed barriers to ability to comply with immunosuppression protocol and follow-up requirements (eg, adequate social support, no severe uncontrolled psychiatric illness)
 - No history of persistent nonadherence
 - No progressive cognitive impairment(1)
 - No severe swallowing dysfunction with recurrent aspiration that is not amenable to therapy (eg, in patient with systemic sclerosis or Sjogren syndrome)(7)

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Footnotes

- A. Transplant may be considered for selected cases of lung-limited cancer (eg, adenocarcinoma in situ, lepidic adenocarcinoma (previously called bronchoalveolar cell carcinoma), or minimally invasive adenocarcinoma) if surgical resection is not otherwise feasible, systemic therapies are contraindicated or have failed, and transplant is thought to be curative.
- B. Patients who required cardiac interventions before transplant are not precluded from lung transplant.(6)
- C. Patients with recent type 2 myocardial infarction (demand ischemia) are not precluded from lung transplant.(1)
- D. Patients with HIV infection and undetectable viral load are not precluded from lung transplant because case series show these patients have similar 5-year survival rates as those without HIV. Coordination of the antiretroviral and immunosuppressive regimens is needed due to risk of drug interactions.(1)
- E. Examples include progressive neuromuscular disease (eg, amyotrophic lateral sclerosis), refractory collagen vascular disease (eg, systemic lupus erythematosus or systemic scleroderma not confined to the lung, rheumatoid arthritis with unstable atlantoaxial instability at risk for spinal cord compression, or severe arthropathy with inability to perform perioperative pulmonary rehabilitation), amyloidosis, or refractory sarcoidosis not confined to the lung.(1)(2)(7) Some patients with cardiac involvement may be considered for combined heart-lung transplant.(7)
- F. The duration of abstinence is discretionary and dependent on the psychosocial evaluation by the transplant center.(1)

Orthostatic hypotension

- Orthostatic hypotension,^{[A][B]} as indicated by **1 or more** of the following(1)(2)(3):
 - Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
 - Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position

References

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3. Fang JC, O'Gara PT. History and physical examination: an evidence-based approach. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. *Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine*. 12th ed. Elsevier; 2022:123-140.

Footnotes

- A. Concomitant measurements of the heart rate are important to measure to help diagnose subtypes of orthostatic hypotension (eg, the lack of a compensatory increase in heart rate is typical of autonomic failure and an exaggerated tachycardia may be reflective of volume depletion). However, the heart rate is not a component of the definition of orthostatic hypotension which relies upon blood pressure alone.(1)(2)(3)
- B. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Respiratory distress

- Respiratory distress, as indicated by **ALL** of the following(1)(2):
 - Patient with **1 or more** of the following:
 - Dyspnea (difficulty breathing)
 - Tachypnea
 - Abnormal breathing pattern (eg, chest retractions)
 - Other evidence of difficulty breathing
 - Evidence of respiratory compromise, as indicated by **1 or more** of the following:
 - Hypoxemia
 - Altered mental status
 - Other evidence of respiratory compromise (eg, respiratory acidosis)

References

1. Braithwaite SA, Wessel AL. Dyspnea. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:194-201.e2.
2. Naureckas ET, Solway J. Disturbances of respiratory function. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:2133-2139.

Social Determinants of Health Assessment

- Risk of poor health outcomes may be increased by the presence of **1 or more** of the following social determinants of health(1)(2)(3)(4):
 - Housing insecurity, as indicated by **1 or more** of the following:
 - Individual or caregiver's current living situation is **1 or more** of the following(5):
 - Does not have own housing (eg, staying in a hotel, shelter, or with others)
 - Has own housing (eg, house, apartment), but at risk of losing it in the future (ie, behind on rent or mortgage)
 - Has own housing (eg, house, apartment), but has lived in 3 or more places in past year
 - Current housing has **1 or more** of the following:
 - Electrical appliances (eg, stove, refrigerator) not working or unavailable
 - Insufficient heating or cooling
 - Insufficient ventilation
 - Lead paint or pipes
 - Mold
 - Pests (eg, bugs) or rodents
 - Smoke detectors not working or unavailable
 - Food insecurity, as indicated by **1 or more** of the following(6):
 - In the past year, individual or caregiver ran out of food and did not have money to buy more food.
 - In the past year, individual or caregiver worried that they would run out of food before they received money to buy more food.
 - Insufficient transportation, as indicated by **1 or more** of the following(7):
 - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of transportation.
 - In the past year, individual or caregiver missed nonmedical activities, work, or could not get things needed for daily living due to lack of transportation.
 - Insufficient utilities, as indicated by **1 or more** of the following(8):
 - Utilities (eg, electricity, water, gas, or oil) are currently shut off or unavailable.
 - In the past year, electric, water, gas, or oil company threatened to shut off services.
 - Personal safety risk, as indicated by **2 or more** of the following(6):
 - Individual is sometimes or frequently physically hurt by another person (including family member).
 - Individual is sometimes or frequently insulted or talked down to by another person (including family member).
 - Individual is sometimes or frequently threatened with physical harm by another person (including family member).
 - Individual is sometimes or frequently screamed or cursed at by another person (including family member).
 - Insufficient dependent care, as indicated by **1 or more** of the following:
 - In the past year, individual or caregiver was unable to work due to lack of dependent care.
 - In the past year, individual or caregiver was unable to work more (additional) hours due to lack of dependent care.
 - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of dependent care.
 - In the past year, individual or caregiver missed nonmedical activities (eg, school, church, social activity) due to lack of dependent care.
 - Depression risk, as indicated by **ALL** of the following:
 - In the past 2 weeks, individual had little interest or pleasure in normal activities on at least several days.
 - In the past 2 weeks, individual felt down, depressed, or hopeless on at least several days.

References

1. Scanlon A, Reinisch C. Social determinants of health. In: Harding MM, Kwong J, Hagler D, Reinisch C, editors. *Lewis's Medical-Surgical Nursing: Assessment and Management of Clinical Problems*. 12th ed. St. Louis, MO: Mosby; 2023:18-20.
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Tachycardia

- Tachycardia,[A][B] as indicated by **1 or more** of the following(1)(2):
 - Heart rate greater than 100 beats per minute in adult or child age 6 years or older
 - Heart rate greater than 115 beats per minute in child 3 to 5 years of age
 - Heart rate greater than 125 beats per minute in child 1 or 2 years of age
 - Heart rate greater than 130 beats per minute in infant 6 to 11 months of age
 - Heart rate greater than 150 beats per minute in infant 3 to 5 months of age
 - Heart rate greater than 160 beats per minute in infant 1 or 2 months of age

References

1. Southmayd GL. Tachycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. *Principles and Practice of Hospital Medicine*. 2nd ed. New York, NY: McGraw-Hill Education; 2017:729-739.
2. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. *The Harriet Lane Handbook: A Manual for Pediatric House Officers*. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Tachycardia absent

- Tachycardia[A][B] absent, as indicated by **1 or more** of the following(1)(2):
 - Heart rate less than or equal to 100 beats per minute in adult or child 6 years or older
 - Heart rate less than or equal to 115 beats per minute in child 3 to 5 years of age
 - Heart rate less than or equal to 125 beats per minute in child 1 or 2 years of age
 - Heart rate less than or equal to 130 beats per minute in infant 6 to 11 months of age
 - Heart rate less than or equal to 150 beats per minute in infant 3 to 5 months of age
 - Heart rate less than or equal to 160 beats per minute in infant 1 or 2 months of age

References

1. Southmayd GL. Tachycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:729-739.
2. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Tachypnea

- Tachypnea,[A][B] as indicated by respiratory rate of **1 or more** of the following(2)(3):
 - Greater than 18 breaths per minute in adult or child 13 years of age or older[A]
 - Greater than 22 breaths per minute in child 6 to 12 years of age[A]
 - Greater than 25 breaths per minute in child 3 to 5 years of age[A]
 - Greater than 30 breaths per minute in child 1 or 2 years of age[A]
 - Greater than 40 breaths per minute in infant 6 to 11 months of age[A]
 - Greater than 45 breaths per minute in infant 3 to 5 months of age[A]
 - Greater than 60 breaths per minute in infant 1 or 2 months of age[A]

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Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. Interpretation of respiratory rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline respiratory rate and whether the respiratory rate is indicative of significant underlying respiratory or other pathology. A mildly increased respiratory rate (eg, 20 to 22 in an adult), absent other indications of serious illness (eg, hypoxemia, metabolic acidosis, decreased tidal volume, pain), may be transitory and not clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachypnea. Whether a respiratory rate above the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment. In addition, the measurement of the respiratory rate is subject to error. A medical textbook states that because there can be significant breath to breath variation in respiratory rate, the rate should be assessed for at least 30 seconds, preferably for a full minute.(1) This same textbook also states that new technologies designed to measure the respiratory rate have not proved to be clinically useful.(1)

Codes

ICD-10 Diagnosis: D86.0, D86.2, E84.0, I27.0, I27.20, I27.21, I27.22, I27.23, I27.24, I27.29, I27.81, I27.83, I27.89, I27.9, J43.0, J43.1, J43.2, J43.8, J43.9, J44.1, J44.9, J47.0, J47.1, J47.9, J84.10, J84.112, J84.81, J84.83, J84.841, J84.842, J84.843, J84.848, J84.89 [Hide]

ICD-10 Procedure: 0BYC0Z0, 0BYC0Z1, 0BYC0Z2, 0BYD0Z0, 0BYD0Z1, 0BYD0Z2, 0BYF0Z0, 0BYF0Z1, 0BYF0Z2, 0BYG0Z0, 0BYG0Z1, 0BYG0Z2, 0BYH0Z0, 0BYH0Z1, 0BYH0Z2, 0BYJ0Z0, 0BYJ0Z1, 0BYJ0Z2, 0BYK0Z0, 0BYK0Z1, 0BYK0Z2, 0BYL0Z0, 0BYL0Z1, 0BYL0Z2, 0BYM0Z0, 0BYM0Z1, 0BYM0Z2 [Hide]

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Last Update: 3/14/2024 8:47:24 AM
Build Number: 28.1.20240313235330.016020